



Titanic Survival Analysis

This analysis uses the Titanic dataset with Python and Plotly to explore which factors influenced survival. Visualizations, preprocessing steps, and key insights are presented in a clean, modern style.

DATA & VIZ

PYTHON + PLOTLY

Project Overview



Dataset: passenger-level records from the Titanic disaster (demographics, ticket info, cabin, and survival outcome).

Columns analyzed:

- Passenger Class (Pclass)
- Gender (Sex)
- Age
- Fare
- Embarked Port
- Cabin (availability)
- Survival (Survived)



Demographics

Age, Sex



Ticket & Fare

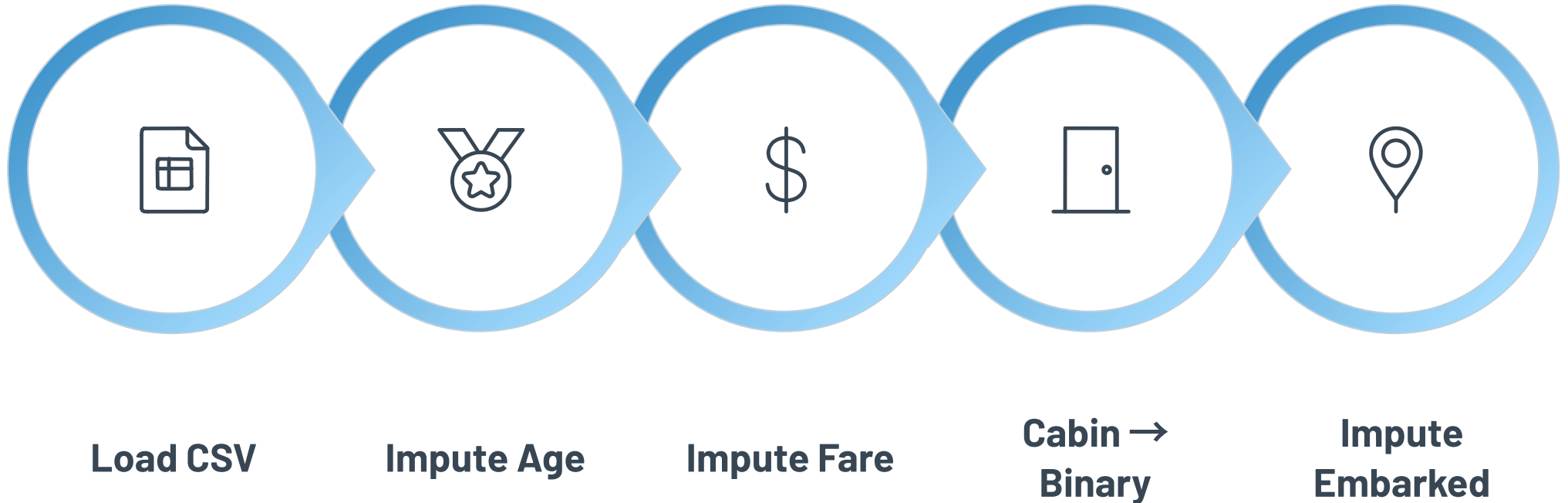
Pclass, Fare, Fare quartiles



Location

Embarked port

Data Cleaning & Preparation



Key steps used before visualization and modeling to ensure clean, comparable groups and enable categorical analyses.

Imputation

Age (median), Fare (median),
Embarked (mode)

Feature Engineering

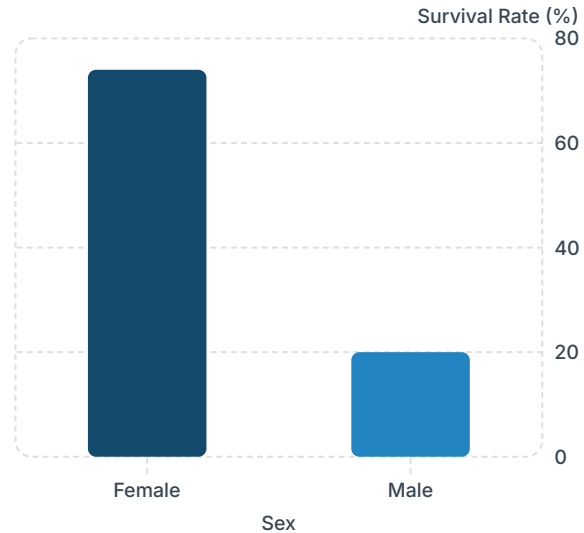
Age groups (child/adult/senior), Fare
quartiles, cabin availability flag

Validation

Sanity checks, distribution
comparison before/after imputation



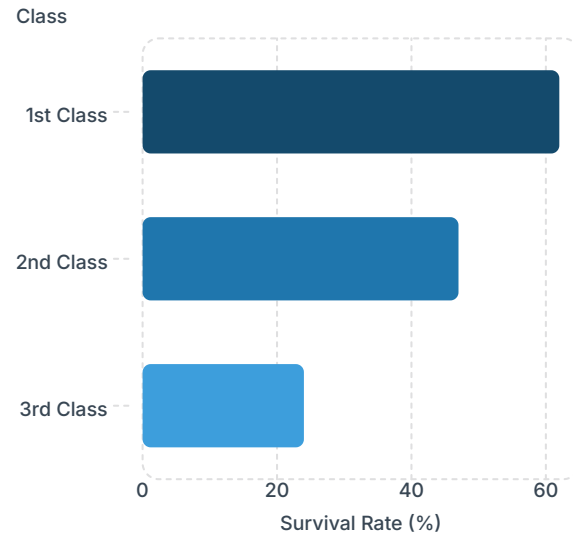
Survival by Gender



Insight: Females had a substantially higher survival rate than males. This pattern is consistent across passenger classes and age groups, reflecting evacuation priorities and social norms at the time.

 Visualization created with Plotly-style column charts and consistent color palette: #75bae6ff and #063E5F.

Survival by Passenger Class



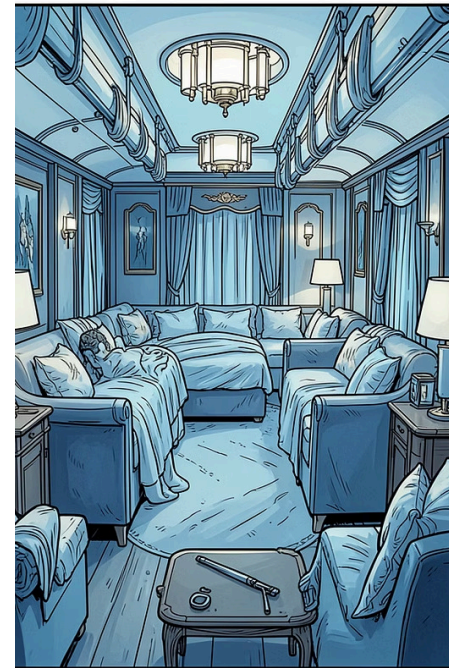
Insight: First-class passengers had the highest survival rate, highlighting socio-economic advantages in access to lifeboats and crew assistance.

Cause

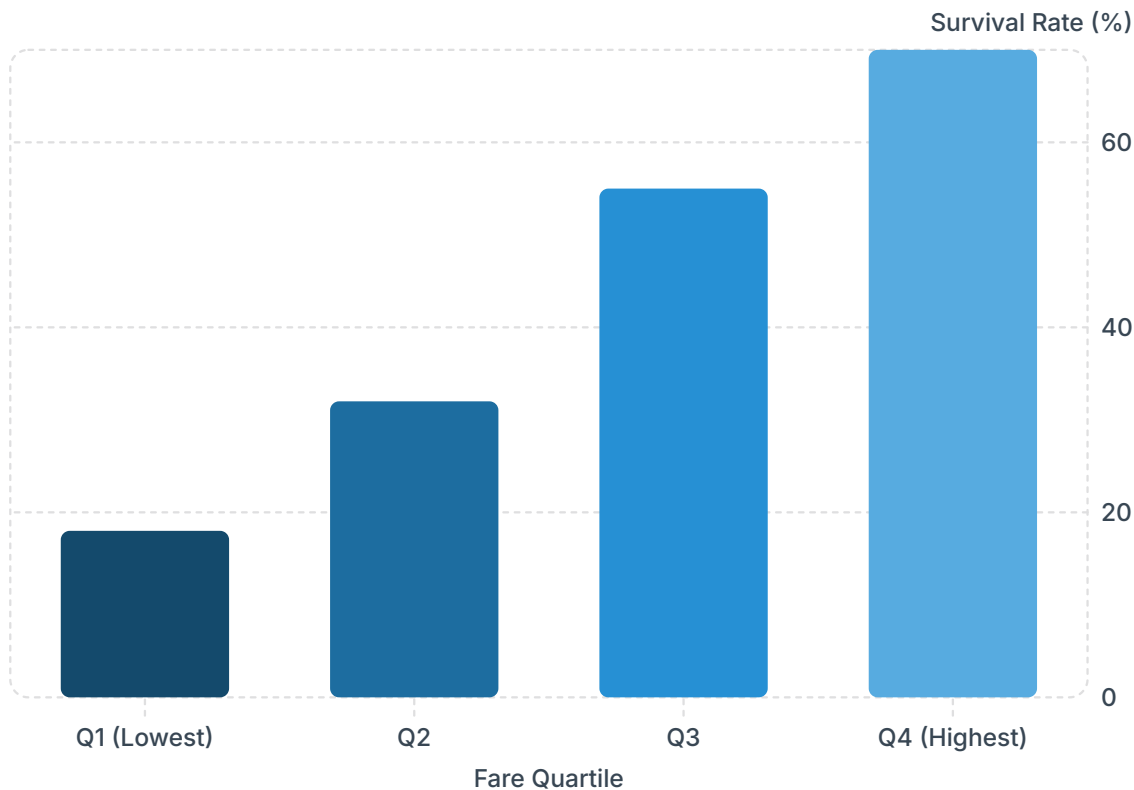
Proximity to decks, priority during evacuation

Impact

Class strongly correlates with survival probability

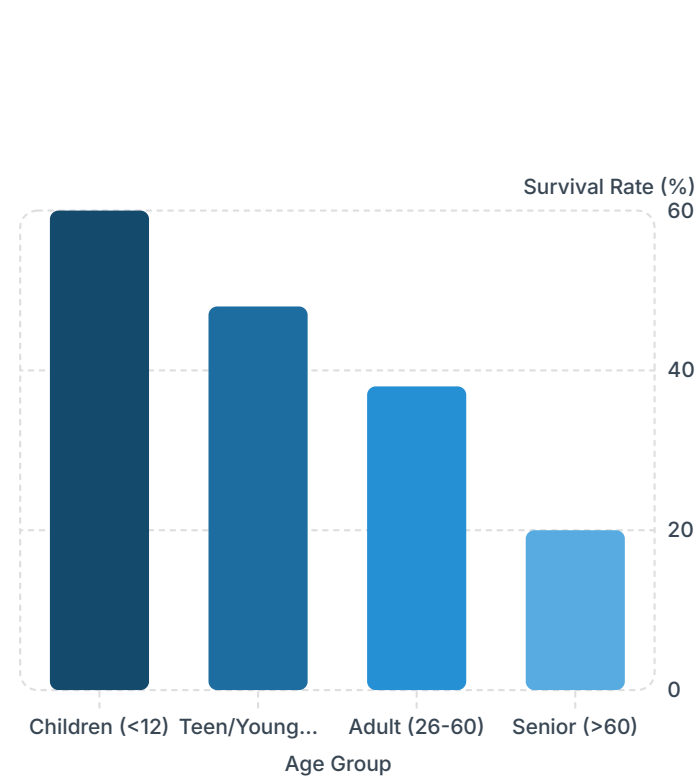


Survival by Fare (Quartiles)



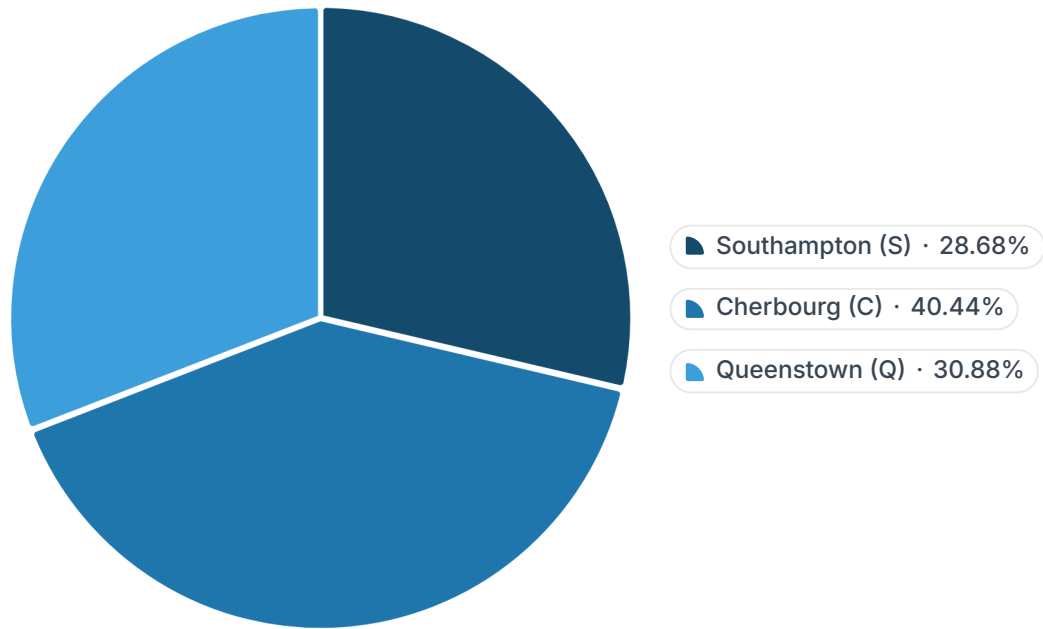
Insight: Higher fares correlate with higher survival rates—fare acts as a proxy for class and cabin location. Consider combining Fare and Pclass for multivariate models.

Survival by Age Group



Insight: Children had higher survival probabilities, reflecting evacuation protocols prioritizing younger passengers. Combine age groups with sex and class for richer insights.

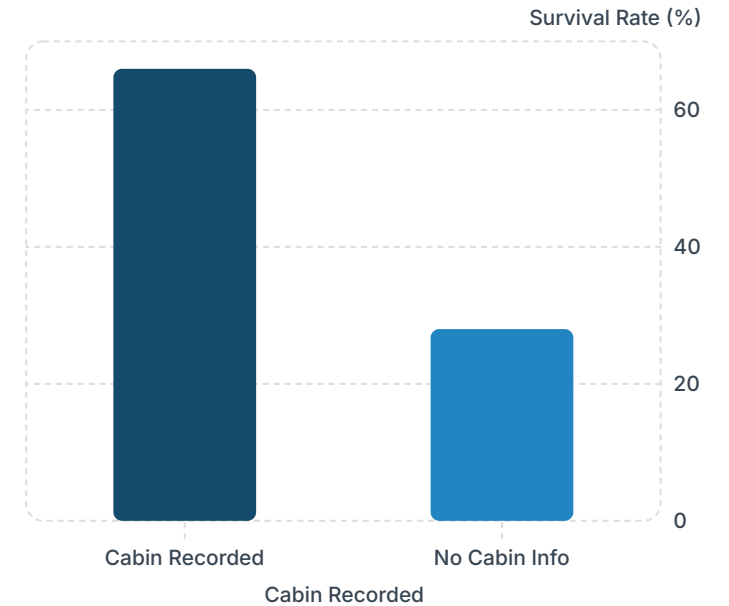
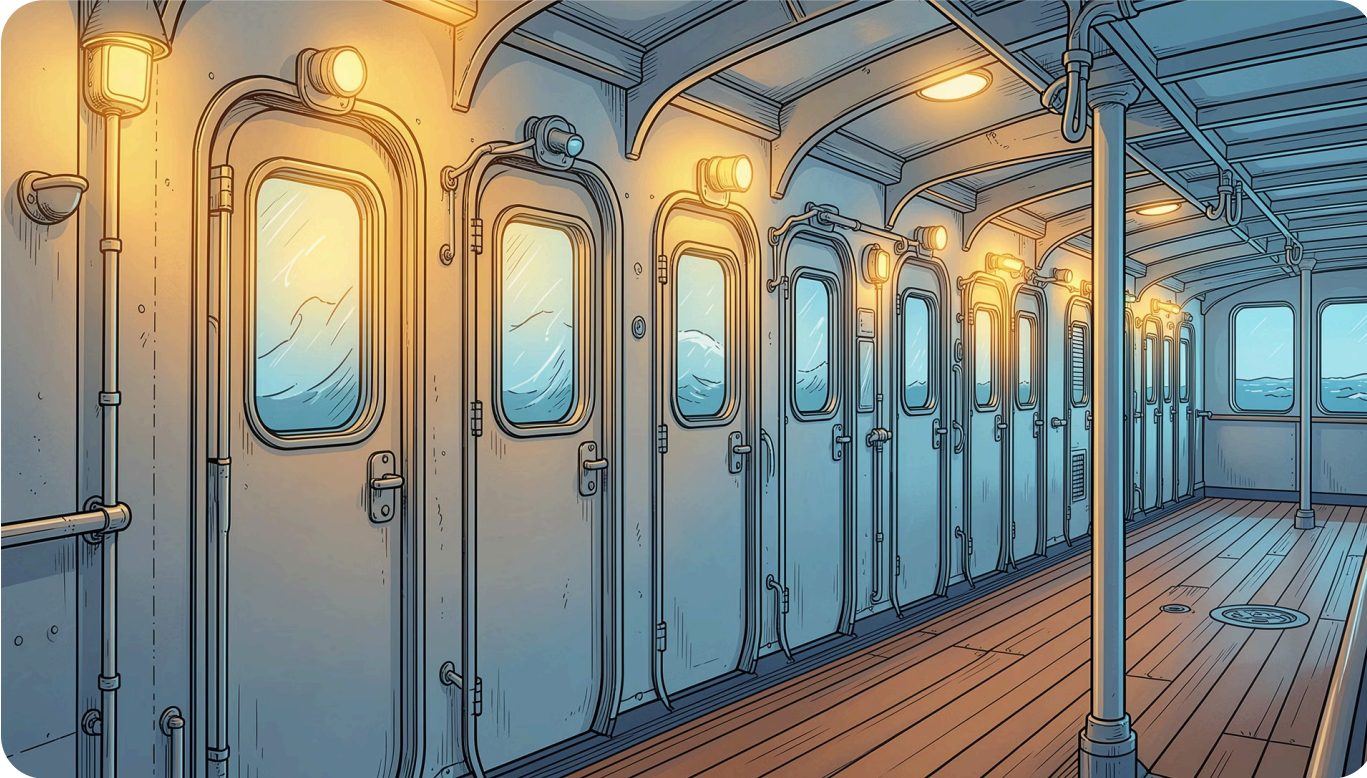
Embarked Port Analysis



Insight: Embarkation port shows differences in survival rates —partly driven by passenger mix (class and fares). Cherbourg passengers had higher average fares and survival.



Cabin Availability Impact



Insight: Passengers with cabin information recorded (generally higher-class tickets) demonstrated substantially higher survival rates. Cabin availability is a useful proxy for socio-economic status.

Final Insights Dashboard

Highest Survival

Female, 1st class, highest fare quartile

Lowest Survival

Male, 3rd class, lowest fare quartile

Top Influencing Factors

Sex, Pclass, Fare, Age group, Cabin availability

Recommendations

Use combined features (sex+pclass+fare+age) for predictive models; consider interaction terms and survival-weighted sampling for class imbalance.

Conclusion

The analysis shows gender, passenger class, and fare were the strongest determinants of survival. Females, children, and first-class passengers had the highest survival probabilities, underscoring how social and economic status influenced outcomes aboard the Titanic.

